

**FEDERAL BOARD OF INTERMEDIATE AND SECONDARY EDUCATION,
ISLAMABAD**

Annual Examination 2026

Chemistry — Class IX | Paper: SSC-I

ANSWER KEY (Version 4)

Syllabus: Ch 1 — Nature of Chemistry | Ch 2 — Matter | Ch 3 — Atomic Structure | Ch 4 — Periodic Table
| Ch 5 — Chemical Bonding | Ch 6 — Stoichiometry | Ch 7 — Electrochemistry | Ch 8 — Energetics | Ch 9
— Chemical Equilibrium

SECTION A — Answers & Explanations (12 × 1 = 12 Marks)

1. B — Industrial chemistry 1

Industrial chemistry deals with the large-scale manufacture of useful chemicals and materials (fertilisers, cement, plastics). Analytical chemistry finds *what* and *how much*; organic chemistry studies carbon compounds; physical chemistry studies energy and rates.

2. C — Salt dissolved in water 1

A **homogeneous** mixture has a uniform composition throughout with no visible boundaries (a true solution). Muddy water, sand in water and oil in water are all **heterogeneous** (components visibly separate).

3. A — 20.2 amu 1

Relative atomic mass = weighted average = $(20 \times 90 + 22 \times 10) \div 100 = (1800 + 220) \div 100 = \mathbf{20.2 \text{ amu}}$. The answer lies close to the more abundant isotope (mass 20).

4. B — position and future path 1

Heisenberg's **uncertainty principle** (1927) states we cannot know both the exact **position** and the future **path (trajectory)** of an electron at the same time. That is why a fixed orbit cannot be drawn, only a probability region (orbital).

5. D — f-block 1

The lanthanides and actinides are the **f-block** (inner transition) elements: their last electrons enter the **f sub-shell**. They are placed at the bottom to keep the table compact.

6. B — polar 1

Dipole-dipole forces act between **polar** molecules: the slightly negative (δ^-) end of one molecule is attracted to the slightly positive (δ^+) end of a neighbour. Non-polar and monatomic species have no permanent dipole.

7. B — 0.4 mol 1

moles = mass $\div$ molar mass = $9.2 \div 23 = \mathbf{0.4 \text{ mol}}$.

8. A — 1.51×10^{23} 1

molecules = moles $\times N_A = 0.25 \times 6.022 \times 10^{23} = \mathbf{1.51 \times 10^{23}}$ molecules.

9. A — +2 1

In MnSO_4 , the sulphate ion SO_4 carries -2 , so Mn must be **+2** to make the compound neutral.

10. B — -120 kJ 1

$\Delta H = (\text{bonds broken}) - (\text{bonds formed}) = 690 - 810 = \mathbf{-120 \text{ kJ}}$. The negative sign means the reaction is exothermic.

11. C — remain constant 1

At equilibrium the forward and reverse rates are equal, so the **concentrations remain constant** (they do not become equal, nor do they fall to zero — both reactions continue).

12. B — increases 1

Down Group 1 the outer electron is farther from the nucleus and more easily lost, so **reactivity increases** (while melting point decreases and density increases).

SECTION B — Short Questions (11 × 3 = 33 Marks)

2(i). Number of moles in 22.2 g of CaCl_2 3

Molar mass $\text{CaCl}_2 = 40 + 2(35.5) = 40 + 71 = 111 \text{ g/mol}$ 1

moles = mass ÷ molar mass = $22.2 \div 111$ 1

moles = 0.2 mol 1

OR — 2(i). Molar mass $\text{N}_2 = 2(14) = 28 \text{ g/mol}$ 1; moles = $5.6 \div 28$ 1 = **0.2 mol** 1.

2(ii). Number of molecules in 0.4 mol of CO_2 3

molecules = moles × N_A 1

= $0.4 \times 6.022 \times 10^{23}$ 1

= **2.409×10^{23} molecules** 1

OR — 2(ii). atoms = moles × N_A 1 = $0.4 \times 6.022 \times 10^{23}$ 1 = **1.807×10^{23} atoms** 1.

2(iii). Molar mass of calcium nitrate, $\text{Ca}(\text{NO}_3)_2$ 3

One $\text{NO}_3 = 14 + 3(16) = 14 + 48 = 62$; there are two $\Rightarrow 2 \times 62 = 124$ 1

$\text{Ca} = 40$ 0.5; total = $40 + 124$ 1

Molar mass = 164 g/mol 0.5

OR — 2(iii). $(\text{NH}_4) = 14 + 4(1) = 18$, $\times 2 = 36$ 1; $\text{SO}_4 = 32 + 4(16) = 96$ 1; formula mass = $36 + 96 = 132 \text{ amu}$ 1.

2(iv). Percentage by mass of nitrogen in NH_4NO_3 3

Molar mass = $14 + 4(1) + 14 + 3(16) = 14 + 4 + 14 + 48 = 80 \text{ g/mol}$ 1

Mass of N in formula = $2 \times 14 = 28$ 0.5

%N = $(28 \div 80) \times 100$ 1

%N = 35% 0.5

OR — 2(iv). Molar mass $\text{MgO} = 24 + 16 = 40$ 1; %O = $(16 \div 40) \times 100$ 1 = **40%** 1.

2(v). Oxidation numbers (let O = -2, H = +1; sum = 0) 3

(a) SO_3 : $\text{S} + 3(-2) = 0 \Rightarrow \text{S} = +6$ 1

(b) CrO_3 : $\text{Cr} + 3(-2) = 0 \Rightarrow \text{Cr} = +6$ 1

(c) P_2O_5 : $2\text{P} + 5(-2) = 0 \Rightarrow 2\text{P} = +10 \Rightarrow \text{P} = +5$ 1

OR — 2(v). (a) N_2O_5 : $2\text{N} + 5(-2) = 0 \Rightarrow \text{N} = +5$ 1; (b) Cl_2O : $2\text{Cl} + (-2) = 0 \Rightarrow \text{Cl} = +1$ 1; (c) HIO_3 : $(+1) + \text{I} + 3(-2) = 0 \Rightarrow \text{I} = +5$ 1.

2(vi). Mole ratio for $2\text{Al} + 3\text{Cl}_2 \rightarrow 2\text{AlCl}_3$ 3

Mole ratio $\text{Al} : \text{Cl}_2 : \text{AlCl}_3 = 2 : 3 : 2$ 1

(a) $\text{Cl}_2 = (3 \div 2) \times 0.4 = 0.6 \text{ mol}$ 1

(b) $\text{AlCl}_3 = (2 \div 2) \times 0.4 = 0.4 \text{ mol}$ 1

OR — 2(vi). $M(\text{CaCO}_3) = 100$; moles = $20 \div 100 = 0.2 \text{ mol}$ 1; ratio 1:1 $\Rightarrow \text{CaO} = 0.2 \text{ mol}$, $M(\text{CaO}) = 56$ 1; mass = $0.2 \times 56 = 11.2 \text{ g}$ 1.

2(vii). Relative atomic mass of element M 3

$A_r = (\text{mass} \times \text{abundance summed}) \div 100$ 0.5

= $(63 \times 69 + 65 \times 31) \div 100 = (4347 + 2015) \div 100$ 1.5

= **$6362 \div 100 = 63.62 \text{ amu}$** 1

OR — 2(vii). $A_r = (10 \times 20 + 11 \times 80) \div 100$ 1.5 = $(200 + 880) \div 100$ 1 = **10.8 amu** 0.5.

2(viii). Heisenberg uncertainty principle 3

It states that it is impossible to determine, at the same time, both the exact **position (location)** and the future **path (trajectory/momentum)** of an electron. 2

Therefore a definite orbit for the electron cannot be drawn; we can only map a region where the electron is **likely** to be found (an orbital). 1

OR — 2(viii). The **quantum mechanical model** is the current model that treats electrons as wave-particle entities and, instead of fixed orbits, gives probability regions where electrons are likely to be found 1. An **orbital** is the region of space around the nucleus where an electron is most likely to be present 1. De Broglie (1924) proposed the **dual nature** of electrons: they can behave both as particles and as waves 1.

2(ix). Radioactivity and the U-238 decay 3

Radioactivity is the spontaneous decay of an unstable nucleus, emitting radiation/particles and changing into another nucleus. 1

Balancing mass and charge: $^{238}_{92}\text{U} \rightarrow ^{234}_{90}\text{Th} + ^4_2\text{He}$ 1

The emitted particle is an **alpha particle** (a doubly positively charged helium nucleus). 1

OR — 2(ix). **Carbon dating** estimates the age of once-living material by comparing the C-14 activity of the sample with that of living tissue 1.5. When $^{14}_6\text{C} \rightarrow ^{14}_7\text{N}$ the proton number **increases from 6 to 7** (a neutron becomes a proton) 1 and a **beta particle** (electron) is emitted 0.5.

2(x). Lanthanides and actinides 3

They are two series of **inner transition** elements. 0.5

They belong to the **f-block**, because their outermost electrons enter (fill) the **f sub-shell**. 1.5

They are shown in two separate rows at the bottom to keep the main table compact (otherwise it would become very wide). 1

OR — 2(x). Down Group 1: (a) reactivity **increases** (outer electron farther from the nucleus, lost more easily) 1; (b) density **increases** 1; (c) melting point **decreases** 1.

2(xi). Dipole-dipole forces 3

They are intermolecular forces of attraction that act between **polar molecules**. 1

In a polar molecule one end is slightly negative (δ^-) and the other slightly positive (δ^+); the δ^- end of one molecule is weakly attracted to the δ^+ end of a neighbouring molecule. 1.5

Example: HCl molecules (also water). 0.5

OR — 2(xi). In **solid** NaCl the ions are locked in fixed positions in the lattice and cannot move, so it does not conduct 1. When **molten** or dissolved, NaCl ionizes and the ions move freely, so it conducts 1.

Aqueous acids conduct because they ionize in water to give freely moving H^+ and negative ions 1.

SECTION C — Long Questions (4 × 5 = 20 Marks)

3(i). Relative atomic mass of uranium + alpha decay 5

(a) Relative atomic mass = the average mass of the isotopes of an element compared with 1/12 the mass of a carbon-12 atom. 0.5

$A_r(\text{U}) = (99.27 \times 238 + 0.72 \times 235 + 0.006 \times 234) \div 100$ 1

$= (23626.26 + 169.2 + 1.404) \div 100$ 1.5

$= 23796.864 \div 100 = 237.97 \text{ amu}$ 0.5

(b) $^{238}_{92}\text{U} \rightarrow ^{234}_{90}\text{Th} + ^4_2\text{He}$ 1; the particle emitted is an **alpha particle** (helium nucleus). 0.5

OR — 3(i). (a) C-12 is the standard: 1 amu is defined as exactly 1/12 the mass of a C-12 atom (C-12 is assigned exactly 12 amu) 1. $A_r(\text{C}) = (98.8 \times 12 + 1.1 \times 13 + 0.009 \times 14) \div 100 = (1185.6 + 14.3 + 0.126) \div 100$ 2 $= 1200.026 \div 100 = 12.00 \text{ amu}$ 0.5. (b) $^{14}_6\text{C} \rightarrow ^{14}_7\text{N} + ^0_{-1}\text{e}$ (beta particle) 1; use: carbon dating / tracing carbon in photosynthesis 0.5.

3(ii). Empirical and molecular formula (M = 88) 5

(a) Take 100 g: C = 54.5 g, H = 9.1 g, O = 36.4 g. Divide each by its atomic mass:

C: $54.5 \div 12 = 4.54$ H: $9.1 \div 1 = 9.1$ O: $36.4 \div 16 = 2.275$ **1.5**

Divide by the smallest (2.275): C = 2.0, H = 4.0, O = 1.0 **1**

Empirical formula = C₂H₄O 0.5

(b) Empirical mass = $2(12) + 4(1) + 16 = 44$ **0.5**; $n = 88 \div 44 = 2$ **1**

Molecular formula = (C₂H₄O)₂ = C₄H₈O₂ 0.5

OR — 3(ii). (a) C: $82.7 \div 12 = 6.89$, H: $17.3 \div 1 = 17.3$ **1**; divide by 6.89 $\Rightarrow$ C = 1, H = 2.51 **1**; $\times 2 =$ **empirical formula = C₂H₅ 1**. (b) Empirical mass = $24 + 5 = 29$; $n = 58 \div 29 = 2$ **1** $\Rightarrow$ **molecular formula = C₄H₁₀ (butane) 1**.

3(iii). ΔH for $N_2 + 3H_2 \rightarrow 2NH_3$ from bond energies 5

Formula: $\Delta H = (\text{sum of bond energies broken}) - (\text{sum of bond energies formed})$ **1**

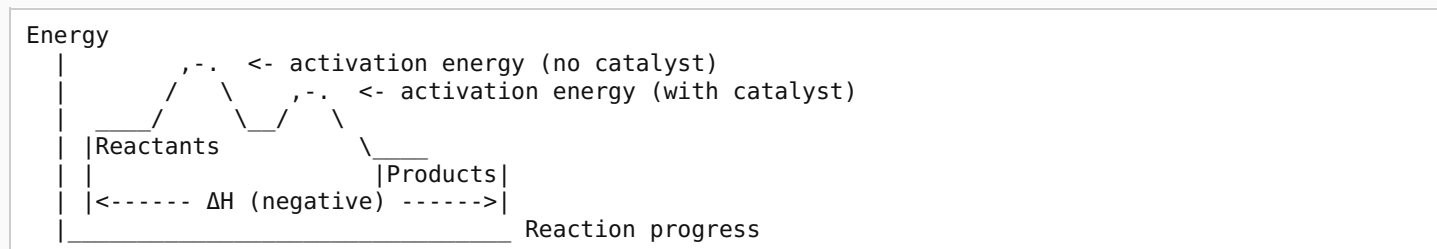
Bonds broken (reactants): $1(N \equiv N) + 3(H-H) = 945 + 3(436) = 945 + 1308 = 2253$ kJ **1.5**

Bonds formed (products): $2 NH_3 = 6(N-H) = 6(391) = 2346$ kJ **1.5**

$\Delta H = 2253 - 2346 = -93$ kJ/mol **0.5**

ΔH is negative, so the reaction is **exothermic**. **0.5**

OR — 3(iii). (a) **Activation energy** = the minimum energy, in addition to the average kinetic energy, that colliding particles must have for a successful (effective) collision **1**. An **effective** collision has enough energy AND correct orientation, so a reaction occurs; an **ineffective** collision lacks sufficient energy or the wrong orientation, so no reaction occurs **2**. (b) Labelled exothermic energy diagram:



ΔH is **negative** (products lie below reactants); the catalyst gives a lower activation-energy path but does not change ΔH **2**.

3(iv). Multi-step stoichiometry: burning 4.4 g propane 5

Molar mass $C_3H_8 = 3(12) + 8(1) = 44$ g/mol.

(a) moles $C_3H_8 = 4.4 \div 44 = 0.1$ mol **1**

(b) $O_2 : C_3H_8 = 5 : 1 \Rightarrow O_2 = 5 \times 0.1 = 0.5$ mol **1**

(c) $CO_2 : C_3H_8 = 3 : 1 \Rightarrow CO_2 = 3 \times 0.1 = 0.3$ mol; $M(CO_2) = 44$; mass = $0.3 \times 44 = 13.2$ g **2**

(d) $H_2O : C_3H_8 = 4 : 1 \Rightarrow H_2O = 0.4$ mol; molecules = $0.4 \times 6.022 \times 10^{23} = 2.409 \times 10^{23}$ **1**

OR — 3(iv). Dot-and-cross diagrams ($\bullet$ = central-atom electron, $\times$ = other-atom electron):

(a) CH ₄	(b) H ₂ O	(c) CO ₂
$ \begin{array}{c} H \\ \\ H - C - H \\ \\ H \end{array} $	$ \begin{array}{c} \bullet \quad \bullet \\ H : O : H \\ \bullet \quad \bullet \end{array} $	$ \begin{array}{c} \bullet \quad \bullet \quad \bullet \quad \bullet \\ O :: C :: O \quad (O = C = O) \\ (:O \text{ each has } 2 \text{ lone pairs}) \end{array} $
C: 4 bonding pairs, 0 lone pairs	O: 2 bonding pairs, 2 lone pairs	C: 4 bonding pairs (two C=O double bonds), 0 lone pairs on C

(a) CH₄: C shares with 4 H $\rightarrow$ **4 bonding pairs, 0 lone pairs** on C **1.5**. (b) H₂O: O forms 2 single bonds $\rightarrow$ **2 bonding pairs and 2 lone pairs** on O **1.5**. (c) CO₂: C forms two double bonds (O=C=O) $\rightarrow$ **4 bonding pairs, 0 lone pairs** on C; each O keeps 2 lone pairs **2**.